Week 1 Project Scope: Product Compare Assistant

Field	Content
Project Name	Product Compare Assistant
Domain	E-commerce / Consumer Electronics (Headphones)

Problem Statement

The rapid growth of e-commerce platforms has provided consumers with many product alternatives, increasing the effort required to compare products and identify meaningful differences between them (Nikhita Vedula et al., 2023).

Product specifications, descriptions, and customer reviews are often distributed across separate pages and presented in different formats, requiring users to manually collect and synthesize information.

Existing search and filtering tools help user narrow product selections but do not provide contextual explanations that answer natural-language comparison questions (Nikhita Vedula et al., 2023).

This project applies AI engineering, specifically retrieval-augmented generation with citation and refusal guardrails, to retrieve evidence from product documents, generate explainable comparisons, and ground every answer in retrieved sources to improve transparency and reliability.

Evidence of the Problem

Evidence 1	Amsl, S., Watson, I., Teller, C., & Wood, S. (2023). Presenting products on websites – the importance of information quality criteria for online shoppers. International Journal of Retail & Distribution Management, 51(9/10). https://doi.org/10.1108/ijrdm-04-2023-0266
Evidence 2	Nikhita Vedula, Collins, M. D, Agichtein, E., & Oleg Rokhlenko. (2023). *Generating explainable product comparisons for online shopping*. https://doi.org/10.1145/3539597.3570489

Initial Related Work

Study / solution 1	Alexa for shopping agentic AI assistant on Amazon
Source / citation	Amazon Staff. (2026, May 13). *Alexa for Shopping: Amazon's AI assistant for personalized shopping*. Amazon News; About Amazon.

	https://www.aboutamazon.com/news/retail/alexa-for-shopping-ai-assistant
Key limitation	Proprietary and closed; the system cannot be independently reproduced or objectively evaluated by researchers.
Study / solution 2	Product question answering in e-commerce: A survey
Source / citation	Deng, Y., Zhang, W., Yu, Q., & Lam, W. (2023). Product question answering in e-commerce: A survey. *Proceedings of the 61St Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 11951–11964. https://doi.org/10.18653/v1/2023.acl-long.667
Key limitation	Existing PQA struggle with answer reliability and trustworthiness; Generation-based methods suffer from hallucination.
Study / solution 3	Meaningful answer generation of e-commerce question-answering
Source / citation	Gao, S., Chen, X., Ren, Z., Zhao, D., & Yan, R. (2021). Meaningful answer generation of e-commerce question-answering. *ACM Transactions on Information Systems*, *39*(2), 1–26. https://doi.org/10.1145/3432689
Key limitation	Focuses on single-product QA rather than product comparison; No explicit evidence attribution or citations

Engineering Gap / Limitation

Gap: Existing comparison generators and commercial assistants are closed or not openly reproducible, and do not combine multi-product comparison with source citation, refusal of unsupported questions, and open evaluation.

Initial Project Objectives

- **O1:** To design a document-grounded AI product comparison assistant for the headphones subcategory of consumer electronics, based on public product metadata, customer reviews, and supporting technical reference documents.
- **O2:** To develop the assistant with hybrid retrieval (keyword plus dense), citation of source products and documents, and guardrails that refuse questions the data cannot support (e.g., current price or stock)
- **O3:** To evaluate the assistant on retrieval quality, answer faithfulness and citation accuracy, refusal correctness, and latency, against BM25-only and dense-only baselines.

Target Users

Online shoppers comparing similar headphones, particularly those who are not technical experts and want plain-language differences backed by evidence.

Data Sources

Source 1	Amazon Reviews 2023, product metadata (descriptions, price, raw image, etc.) for one electronics subcategory. URL: McAuley-Lab/Amazon-Reviews-2023 · Datasets at Hugging Face
Source 2	Amazon Reviews 2023, user reviews (ratings, text, helpfulness votes, etc.) Same URL as Source 1.
Source 3	Olive, S. E. (2021). The perception and measurement of headphone sound quality: What do listeners prefer? *Acoustics Today*, *18*(1), 58. https://doi.org/10.1121/at.2022.18.1.58
Source 4	Cotton, D. (2021). *Headphone essentials 3: Basics of headphone types and tech* (Version 1.31). Daystar Visions. http://daystarvisions.com/Music/HE3-Basics_of_Headphone_Types_and_Tech.pdf

Similar to EduAdvisor

Document-grounded Q&A using RAG; answers cite official or source passages; guardrails and refusal for out-of-scope questions; evaluation of answer quality, groundedness, latency, and safety.

Different from EduAdvisor

- Semi-structured product data (metadata fields plus free-text reviews) instead of policy PDFs.
- Multi-item comparison: the assistant must retrieve and contrast several products, not look up one rule.
- Review text can be subjective, conflicting, or manipulated, so evidence must be weighed and attributed.
- Data is a 2023 snapshot, so the assistant must refuse current price and stock questions.
- Possible extension toward agentic behaviour (e.g., a comparison tool) in later topics.

Responsible AI Risks

- **Risk 1: Hallucination.** The model invents or misstates a specification or compatibility claim, leading to a wrong purchase. *Control:* answer only from retrieved passages, show source product IDs, refuse when evidence is missing.
- **Risk 2: Review bias and manipulation.** Fake, promotional, or skewed reviews distort the comparison. *Control:* attribute opinions to reviews, report review counts, and show conflicting views rather than a single verdict.
- **Risk 3: Unfair coverage and stale data.** Some brands or subcategories have far more data than others, and 2023 data may be outdated. *Control:* report coverage per brand, state the data date, and decline price or availability questions.